

PRACTICAL-4

CODE:

```
GNU nano 7.2 multiple_children.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid1, pid2;

    pid1 = fork();

    if (pid1 == 0)
    {
        printf("Child 1\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        sleep(2);

        printf("Child 1 Finished\n");
        exit(0);
    }

    pid2 = fork();

    if (pid2 == 0)
    {
        printf("Child 2\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());

        sleep(4);

        printf("Child 2 Finished\n");
        exit(0);
    }

    printf("\nParent Process\n");
    printf("Parent PID : %d\n", getpid());

    wait(NULL);
    printf("One child completed using wait().\n");

    waitpid(pid2, NULL, 0);
    printf("Second child completed using waitpid().\n");
}
```

```
GNU nano 7.2 zombie_fixed.c
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("PID : %d\n", getpid());

        exit(0);
    }
    else
    {
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());

        wait(NULL);

        printf("Child collected successfully.\n");
        printf("No Zombie Process.\n");
    }

    return 0;
}
```

OUTPUT:

1)

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-04$ ./multiple_children
Parent Process
Child 1
Parent PID : 1808
PID : 1809
PPID : 1808
Child 2
PID : 1810
PPID : 1808
Child 1 Finished
One child completed using wait().
Child 2 Finished
Second child completed using waitpid().
All child processes completed.
```

2)

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-04$ ./zombie_process
Parent Process
PID : 1830
Sleeping for 20 seconds...
Child Process
PID : 1831
Child exiting...
^N^N^TParent exiting.
```

3)

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Practical/Practical-04$ ./zombie_fixed
Parent Process
PID : 2535
Child Process
PID : 2536
Child collected successfully.
No Zombie Process.
```